

FDS

Fundamentals of Distributed systems

ASSIGNMENT – Q1

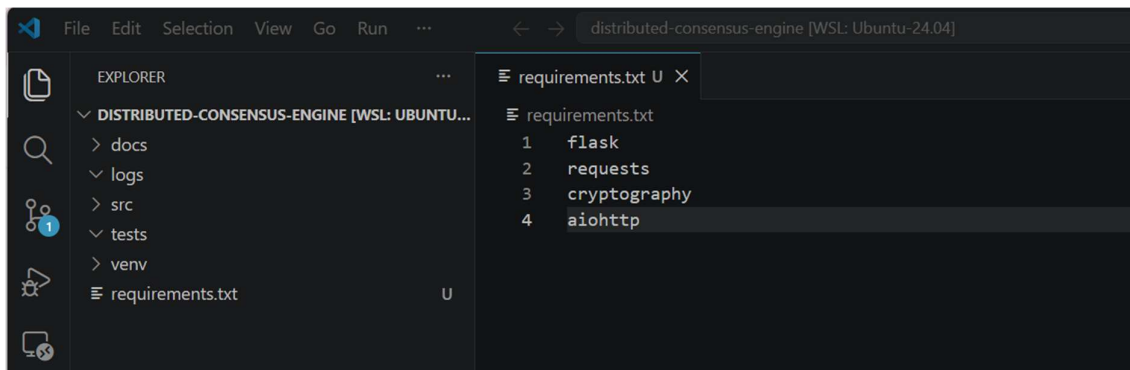
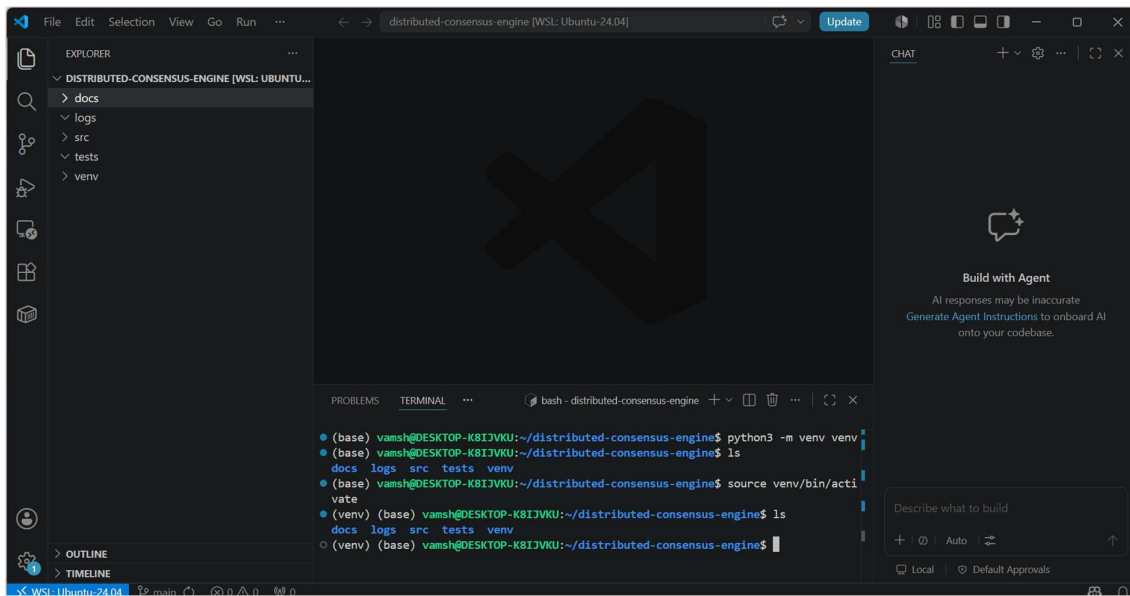
Resilient State Machine Replication: Paxos & PBFT

SUBMITTED BY

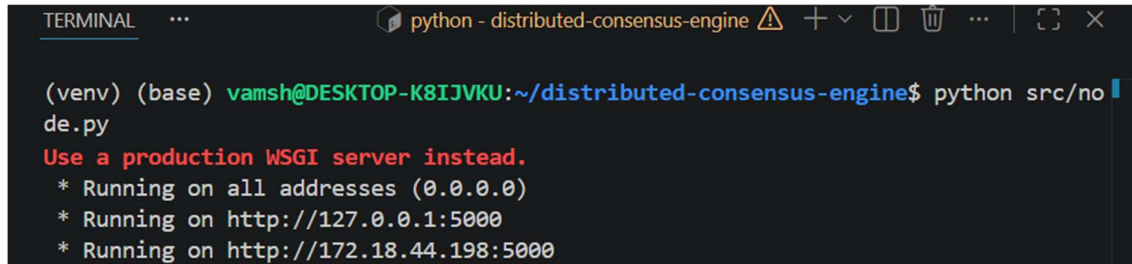
N A M E Vamshikumar Bejjarapu	R O L L N U M B E R G25AI1066
GitHub Repository	https://github.com/Vamshi13212/distributed-consensus-engine
Video Link	https://drive.google.com/file/d/12-pTUUUFmvT2rizmV5kLaYbk394e4FGa/view?usp=sharing

The below screenshot to show that repository was successfully initialized and linked to GitHub. The project directory structure was prepared for subsequent development phases

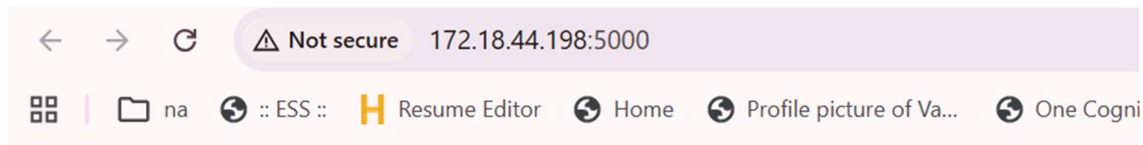
```
(base) vamsh@DESKTOP-K8IJVKU: /mnt/c/Users/Vamsh$ cd ~
(base) vamsh@DESKTOP-K8IJVKU: ~$ mkdir distributed-consensus-engine
(base) vamsh@DESKTOP-K8IJVKU: ~$ cd distributed-consensus-engine
(base) vamsh@DESKTOP-K8IJVKU: ~/distributed-consensus-engine$ pwd
/home/vamsh/distributed-consensus-engine
(base) vamsh@DESKTOP-K8IJVKU: ~/distributed-consensus-engine$ git init
Initialized empty Git repository in /home/vamsh/distributed-consensus-engine/.git/
(base) vamsh@DESKTOP-K8IJVKU: ~/distributed-consensus-engine$ git remote add origin git@github.com:Vamshi13212/distributed-consensus-engine.git
(base) vamsh@DESKTOP-K8IJVKU: ~/distributed-consensus-engine$ git remote -v
origin  git@github.com:Vamshi13212/distributed-consensus-engine.git (fetch)
origin  git@github.com:Vamshi13212/distributed-consensus-engine.git (push)
(base) vamsh@DESKTOP-K8IJVKU: ~/distributed-consensus-engine$
```



This figure shows the installation of required Python libraries, including Flask, Requests, Cryptography, and AIOHTTP. The screenshots also demonstrate the implementation of the first distributed node using Flask and the successful execution of the node service on port 5000.



```
python - distributed-consensus-engine
(venv) (base) vamsh@DESKTOP-K8IJVKU:~/distributed-consensus-engine$ python src/node.py
Use a production WSGI server instead.
* Running on all addresses (0.0.0.0)
* Running on http://127.0.0.1:5000
* Running on http://172.18.44.198:5000
```



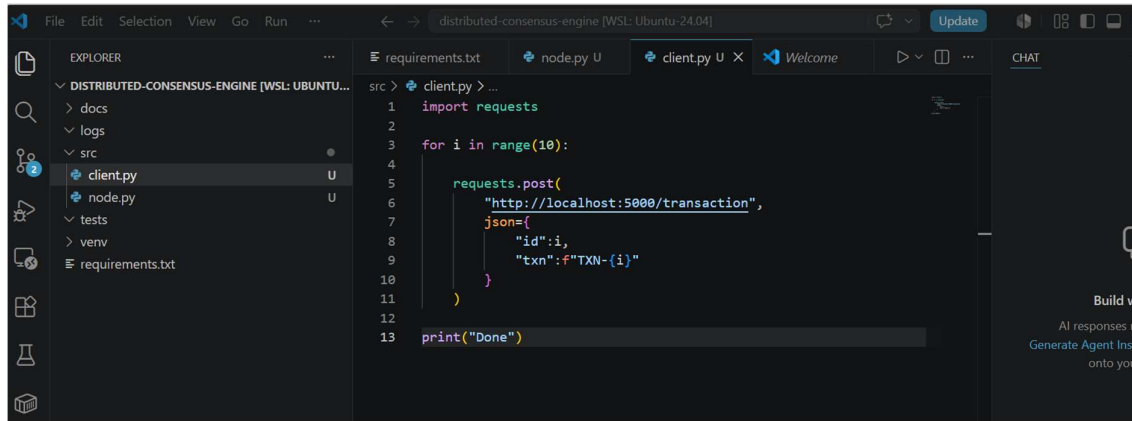
Node Running

Create Client

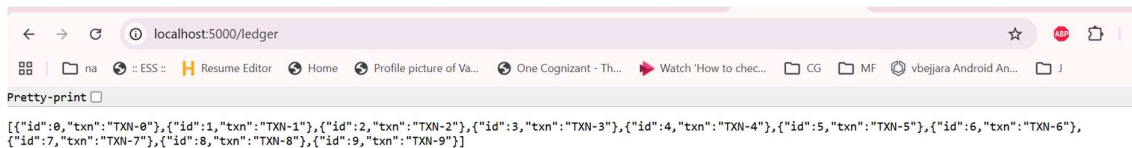
This figure presents the development of the client application responsible for generating and submitting transactions to the distributed node. The node service was executed and verified through browser access, while the client generated multiple transaction requests.

Observation: The client successfully communicated with the server using HTTP requests.

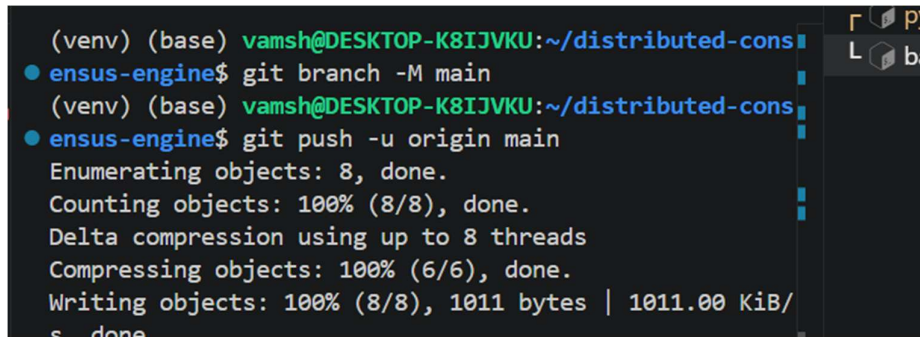
Outcome: Basic client-server communication was validated, confirming the correctness of the REST API implementation and transaction handling logic.



```
1 import requests
2
3 for i in range(10):
4
5     requests.post(
6         "http://localhost:5000/transaction",
7         json={
8             "id":i,
9             "txn":f"TXN-{i}"
10        }
11    )
12
13 print("Done")
```

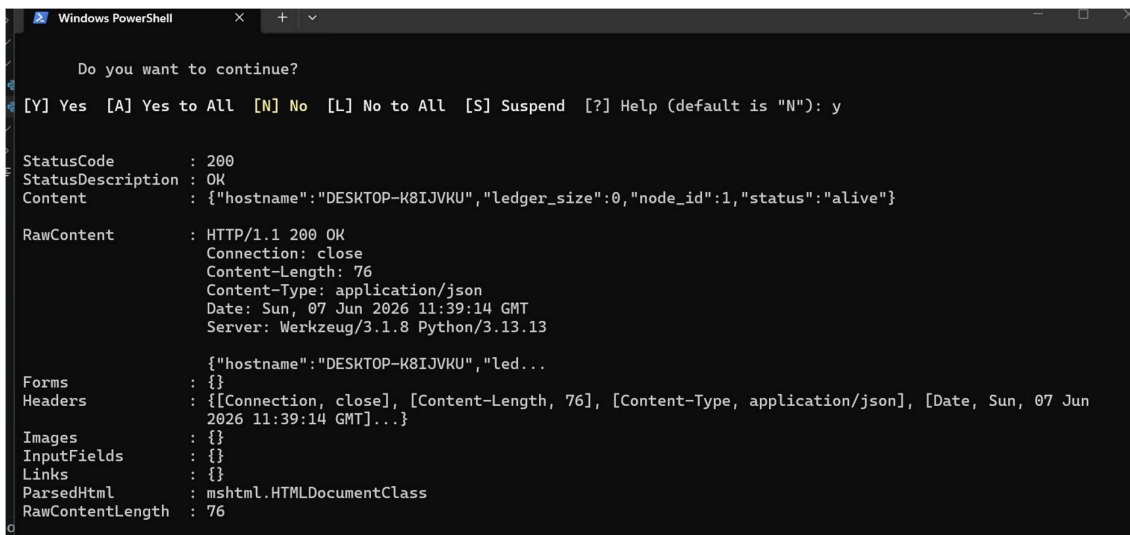
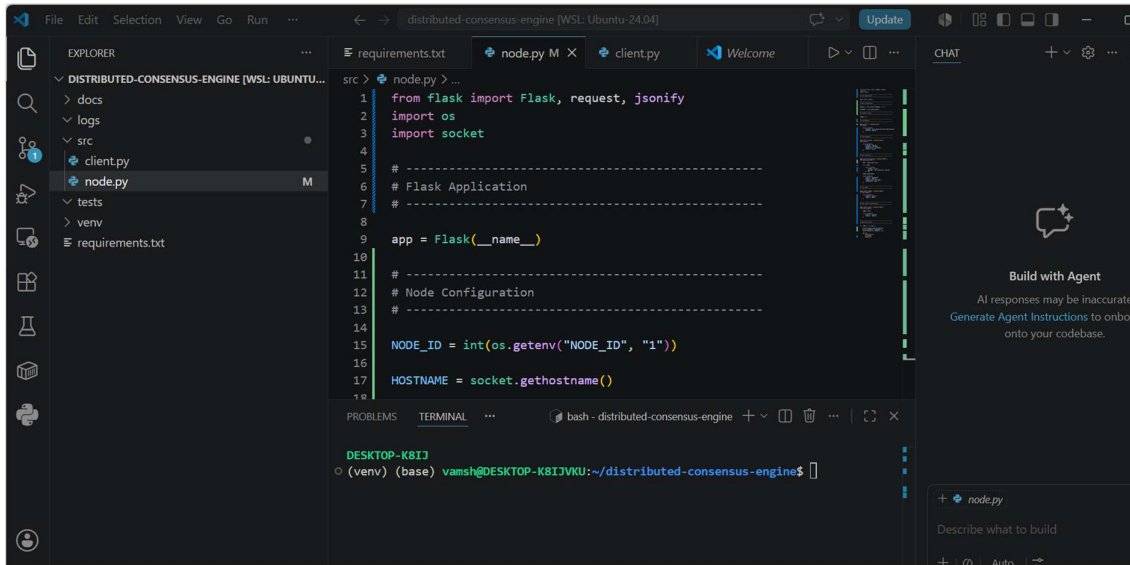


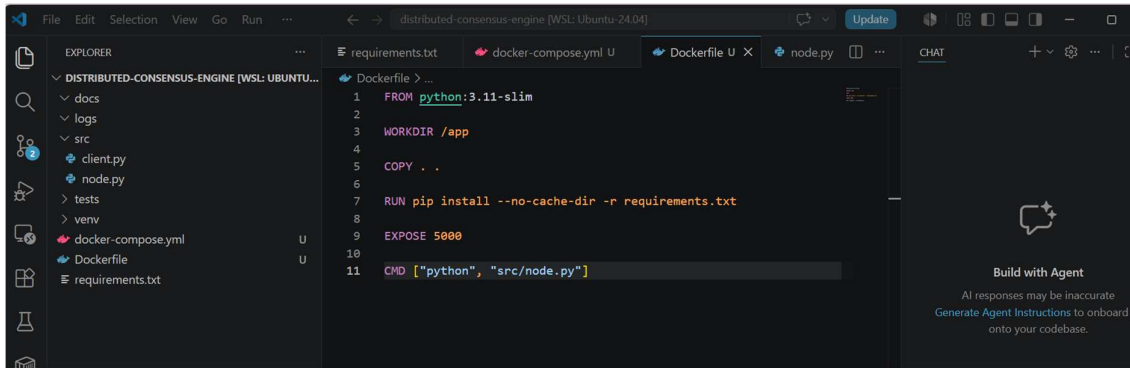
```
[{"id":0,"txn":"TXN-0"}, {"id":1,"txn":"TXN-1"}, {"id":2,"txn":"TXN-2"}, {"id":3,"txn":"TXN-3"}, {"id":4,"txn":"TXN-4"}, {"id":5,"txn":"TXN-5"}, {"id":6,"txn":"TXN-6"}, {"id":7,"txn":"TXN-7"}, {"id":8,"txn":"TXN-8"}, {"id":9,"txn":"TXN-9"}]
```



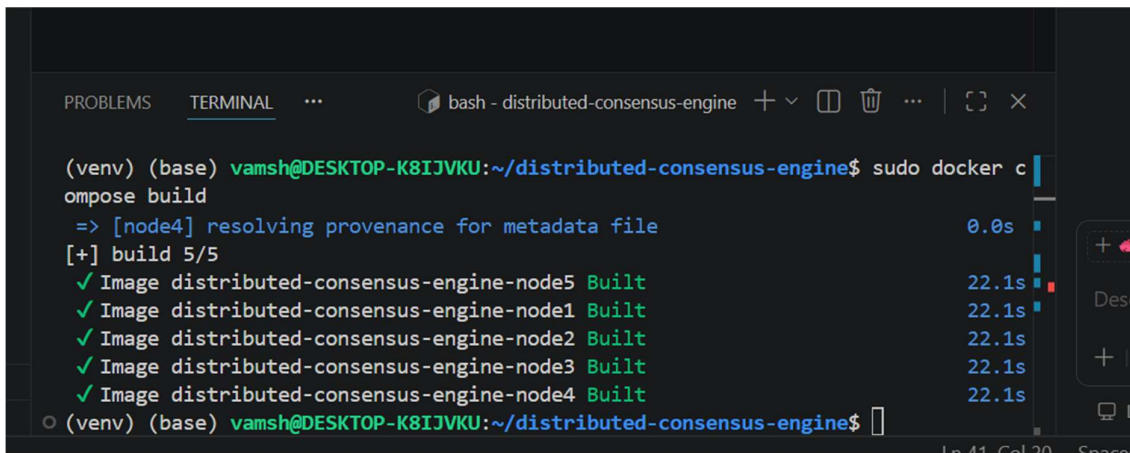
```
(venv) (base) vamsh@DESKTOP-K8IJVKU:~/distributed-consensus-engine$ git branch -M main
(venv) (base) vamsh@DESKTOP-K8IJVKU:~/distributed-consensus-engine$ git push -u origin main
Enumerating objects: 8, done.
Counting objects: 100% (8/8), done.
Delta compression using up to 8 threads
Compressing objects: 100% (6/6), done.
Writing objects: 100% (8/8), 1011 bytes | 1011.00 KiB/s, done
```

Before Leader Election and Paxos. We first make sure all 5 nodes can communicate.

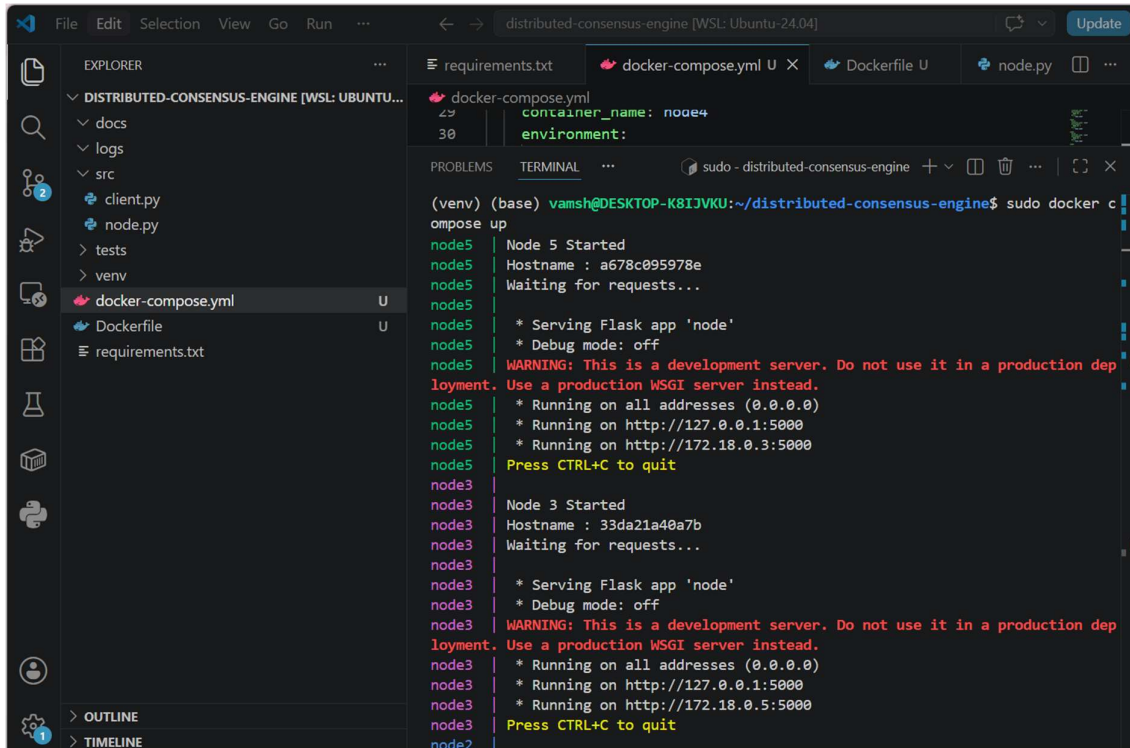




Build Docker Image



docker compose up and running, all nodes are started



```
requirements.txt
docker-compose.yml
Dockerfile
node.py

docker-compose.yml
container_name: node4
environment:

PROBLEMS
TERMINAL
sudo - distributed-consensus-engine

(venv) (base) vamsh@DESKTOP-K8IJVKU:~/distributed-consensus-engine$ sudo docker c
ompose up
node5 | Node 5 Started
node5 | Hostname : a678c095978e
node5 | Waiting for requests...
node5 |
node5 | * Serving Flask app 'node'
node5 | * Debug mode: off
node5 | WARNING: This is a development server. Do not use it in a production dep
loyment. Use a production WSGI server instead.
node5 | * Running on all addresses (0.0.0.0)
node5 | * Running on http://127.0.0.1:5000
node5 | * Running on http://172.18.0.3:5000
node5 | Press CTRL+C to quit
node3 |
node3 | Node 3 Started
node3 | Hostname : 33da21a40a7b
node3 | Waiting for requests...
node3 |
node3 | * Serving Flask app 'node'
node3 | * Debug mode: off
node3 | WARNING: This is a development server. Do not use it in a production dep
loyment. Use a production WSGI server instead.
node3 | * Running on all addresses (0.0.0.0)
node3 | * Running on http://127.0.0.1:5000
node3 | * Running on http://172.18.0.5:5000
node3 | Press CTRL+C to quit
node2 |
```

```
(venv) (base) vamsh@DESKTOP-K8IJVKU:~/distributed-consensus-engine$ sudo docker ps
[sudo] password for vamsh:
CONTAINER ID   IMAGE                                COMMAND                  CREATED        STATUS        PORTS                                     NAMES
9e1f2af8b2e6   distributed-consensus-engine-node2   "python src/node.py"    2 minutes ago Up 2 minutes  0.0.0.0:5002->5000/tcp, [::]:5002->5000/tcp node2
33da21a40a7b   distributed-consensus-engine-node3   "python src/node.py"    2 minutes ago Up 2 minutes  0.0.0.0:5003->5000/tcp, [::]:5003->5000/tcp node3
c1bc3882c668   distributed-consensus-engine-node1   "python src/node.py"    2 minutes ago Up 2 minutes  0.0.0.0:5001->5000/tcp, [::]:5001->5000/tcp node1
1eb455028f49   distributed-consensus-engine-node4   "python src/node.py"    2 minutes ago Up 2 minutes  0.0.0.0:5004->5000/tcp, [::]:5004->5000/tcp node4
a678c095978e   distributed-consensus-engine-node5   "python src/node.py"    2 minutes ago Up 2 minutes  0.0.0.0:5005->5000/tcp, [::]:5005->5000/tcp node5
(venv) (base) vamsh@DESKTOP-K8IJVKU:~/distributed-consensus-engine$
```

This proves Docker networking works.

```
(venv) (base) vamsh@DESKTOP-K8IJVKU:~/distributed-consensus-engine$ curl http://localhost:5001/status
{"hostname":"c1bc3882c668","ledger_size":0,"node_id":1,"status":"alive"}
(venv) (base) vamsh@DESKTOP-K8IJVKU:~/distributed-consensus-engine$ curl http://localhost:5002/status
{"hostname":"9e1f2af8b2e6","ledger_size":0,"node_id":2,"status":"alive"}
(venv) (base) vamsh@DESKTOP-K8IJVKU:~/distributed-consensus-engine$ curl http://localhost:5003/status
{"hostname":"33da21a40a7b","ledger_size":0,"node_id":3,"status":"alive"}
(venv) (base) vamsh@DESKTOP-K8IJVKU:~/distributed-consensus-engine$ curl http://localhost:5004/status
{"hostname":"1eb455028f49","ledger_size":0,"node_id":4,"status":"alive"}
(venv) (base) vamsh@DESKTOP-K8IJVKU:~/distributed-consensus-engine$ curl http://localhost:5004/status
{"hostname":"1eb455028f49","ledger_size":0,"node_id":4,"status":"alive"}
(venv) (base) vamsh@DESKTOP-K8IJVKU:~/distributed-consensus-engine$
```

Send Transaction to Node1


```
(base) vamsh@DESKTOP-K8IJVKU:/mnt/c/Users/Vamsh$ curl -X POST http://localhost:5001/transaction \
-H "Content-Type: application/json" \
-d '{"txn":"A pays B 100"}'
{"ledger_size":1,"node_id":1,"status":"committed","transaction":{"txn":"A pays B 100"}}
(base) vamsh@DESKTOP-K8IJVKU:/mnt/c/Users/Vamsh$
```

Check Ledger

Pretty-print ☐

```
{"ledger":[{"txn":"A pays B 100"}],"node_id":1}
```

Verify Node Isolation

Pretty-print ☐

```
{"ledger":[],"node_id":2}
```

```
PROBLEMS  TERMINAL  ...  bash - distributed-consensus-engine
(venv) (base) vamsh@DESKTOP-K8IJVKU:~/distributed-consensus-engine$ sudo docker c
ompose up
Container node4 Stopping
Container node3 Stopping
Container node2 Stopping
Container node1 Stopped
node1 exited with code 137
Container node2 Stopped
node2 exited with code 137
Container node3 Stopped
node3 exited with code 137
Container node5 Stopped
node5 exited with code 137
Container node4 Stopped
node4 exited with code 137
○ (venv) (base) vamsh@DESKTOP-K8IJVKU:~/distributed-consensus-engine$
```

Leader Election (Bully Algorithm)

```

src > node.py > ...
1  from flask import Flask, request, jsonify
2
3  import os
4  import socket
5  import requests
6  import threading
7  import time
8
9  # =====
10 # Flask Application
11 # =====
12
13 app = Flask(__name__)
14
15 # =====
16 # Node Configuration
17 # =====
18
19 NODE_ID = int(os.getenv("NODE_ID", "1"))
20
21 HOSTNAME = socket.gethostname()
22
23 # =====
24 # Cluster Configuration
25 # =====
26
27 nodes = {
28     1: "http://node1:5000",
29     2: "http://node2:5000",
30     ...

```

```

(vamsh@DESKTOP-K81JVKU:~/distributed-consensus-engine$ docker compose up
unable to get image 'distributed-consensus-engine-node5': permission denied while trying to connect to the docker API at unix:///var/run/docker.sock
(vamsh@DESKTOP-K81JVKU:~/distributed-consensus-engine$ sudo docker compose up
[+] up 6/6
✓ Network distributed-consensus-engine_default Created
✓ Container node3 Created
✓ Container node2 Created
✓ Container node1 Created
✓ Container node5 Created
✓ Container node4 Created
Attaching to node1, node2, node3, node4, node5
node4 |
node4 | =====
node4 | Node 4 Started
node4 | Hostname : 344147e54bd3
node4 | Starting Leader Election Service
node4 | =====
node4 | * Serving Flask app 'node'
node4 | * Debug mode: off
node5 |
node4 | WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
node1 |
node3 | =====
node3 | Hostname : dfcc75685f9d
node5 | =====
node5 | Node 5 Started
node5 | Hostname : 9be6e1dc6be4

```

Check Leader

```

• (venv) (base) vamsh@DESKTOP-K8IJVKU:~/distributed-consensus-engine$ curl http://localhost:5002/leader
{"leader_id":5}
• (venv) (base) vamsh@DESKTOP-K8IJVKU:~/distributed-consensus-engine$ curl http://localhost:5001/leader
{"leader_id":5}
• (venv) (base) vamsh@DESKTOP-K8IJVKU:~/distributed-consensus-engine$ curl http://localhost:5003/leader
{"leader_id":5}
• (venv) (base) vamsh@DESKTOP-K8IJVKU:~/distributed-consensus-engine$ curl http://localhost:5004/leader
{"leader_id":5}
• (venv) (base) vamsh@DESKTOP-K8IJVKU:~/distributed-consensus-engine$ curl http://localhost:5005/leader
{"leader_id":5}
○ (venv) (base) vamsh@DESKTOP-K8IJVKU:~/distributed-consensus-engine$ █

```

Simulate Leader Failure

```

• (venv) (base) vamsh@DESKTOP-K8IJVKU:~/distributed-consensus-engine$ sudo docker stop node5
[sudo] password for vamsh:
node5
• (venv) (base) vamsh@DESKTOP-K8IJVKU:~/distributed-consensus-engine$ curl http://localhost:5001/leader
{"leader_id":4}
○ (venv) (base) vamsh@DESKTOP-K8IJVKU:~/distributed-consensus-engine$ █

```

Recover Leader

```

Failed to start containers: node5
• (venv) (base) vamsh@DESKTOP-K8IJVKU:~/distributed-consensus-engine$ sudo docker start node5
node5
• (venv) (base) vamsh@DESKTOP-K8IJVKU:~/distributed-consensus-engine$ curl http://localhost:5001/leader
{"leader_id":5}
○ (venv) (base) vamsh@DESKTOP-K8IJVKU:~/distributed-consensus-engine$ █

```

Heartbeats : Detect leader failure via heartbeats and elect a new leader automatically

```

• (venv) (base) vamsh@DESKTOP-K8IJVKU:~/distributed-consensus-engine$ sudo docker stop node5
[sudo] password for vamsh:
node5

```

```

[Node 1] Leader timeout detected
[Node 1] New Leader Elected: Node 4

```

localhost:5001/leader

na :: ESS :: Resume Editor Home P

Pretty-print ☐

```
{"leader_id":4}
```

```
(venv) (base) vamsh@DESKTOP-K8IJVKU:~/distributed-consensus-engine$ sudo docker start node5
node5
(venv) (base) vamsh@DESKTOP-K8IJVKU:~/distributed-consensus-engine$
```

localhost:5001/leader

na :: ESS :: Resume Editor Home Profile picture of Va... One Cogniz

Pretty-print ☐

```
{"leader_id":5}
```

Phase 12 : Paxos implementation

localhost:5001/leader

na :: ESS :: Resume Editor Home Profile picture of Va...

Pretty-print ☐

```
{"leader_id":5}
```

```
(base) vamsh@DESKTOP-K8IJVKU:~/distributed-consensus-engine$ source /home/vamsh/distributed-consensus-engine/venv/bin/activate
(venv) (base) vamsh@DESKTOP-K8IJVKU:~/distributed-consensus-engine$ curl http://localhost:5001/leader
{"leader_id":5}
(venv) (base) vamsh@DESKTOP-K8IJVKU:~/distributed-consensus-engine$ curl -X POST http://localhost:5005/propose_transaction \
-H "Content-Type: application/json" \
-d '{"txn":"A pays B 100"}'
{"leader":5,"status":"committed"}
(venv) (base) vamsh@DESKTOP-K8IJVKU:~/distributed-consensus-engine$ curl http://localhost:5005/ledger
{"leader_id":5,"ledger":[{"txn":"A pays B 100"}],"node_id":5}
(venv) (base) vamsh@DESKTOP-K8IJVKU:~/distributed-consensus-engine$
```

Phase 13: PBFT Cryptographic Signatures

Node1



Digitally signs message



Node2 verifies signature



Accept only if valid

The screenshot shows a VS Code editor window titled 'distributed-consensus-engine [WSL: Ubuntu-24.04]'. The Explorer sidebar on the left shows a project structure with folders 'docs', 'logs', 'src', and 'tests', and files 'client.py', 'crypto_utils.py', 'heart beat failure', 'node.py', 'paxos', 'split-brain scenario', 'docker-compose.yml', 'Dockerfile', and 'requirements.txt'. The 'src' folder is expanded, showing 'crypto_utils.py' selected. The main editor displays the 'verify_signature' function in 'crypto_utils.py'. The function takes 'public_key', 'message', and 'signature' as arguments and returns a boolean. It uses 'public_key.verify' to check the signature against the message and a SHA256 hash. The function is wrapped in a try-except block to handle exceptions.

```
34 # =====  
35 # Verify Signature  
36 # =====  
37  
38 def verify_signature(  
39     public_key,  
40     message,  
41     signature  
42 ):  
43  
44     try:  
45  
46         public_key.verify(  
47             signature,  
48             message.encode(),  
49             padding.PKCS1v15(),  
50             hashes.SHA256()  
51         )  
52  
53         return True  
54  
55     except Exception:  
56  
57         return False
```

```
nodes exited with code 137  
• (venv) (base) vamsh@DESKTOP-K8IJVKU:~/distributed-consensus-engine$ curl http://localhost:5001/pbft_test  
{"message":"TEST_PBFT","signature":"186ac14ee609561cc57b49a588d0854c96138630bbe2c4a356"}  
○ (venv) (base) vamsh@DESKTOP-K8IJVKU:~/distributed-consensus-engine$ ch Detach
```

```
• (venv) (base) vamsh@DESKTOP-K8IJVKU:~/distributed-consensus-engine$ curl http://localhost:5001/pbft_test  
{"message":"TEST_PBFT","signature":"186ac14ee609561cc57b49a588d0854c96138630bbe2c4a356"}  
• (venv) (base) vamsh@DESKTOP-K8IJVKU:~/distributed-consensus-engine$ curl http://localhost:5001/pbft_test  
{"message":"TEST_PBFT","signature":"186ac14ee609561cc57b49a588d0854c96138630bbe2c4a356"}  
• (venv) (base) vamsh@DESKTOP-K8IJVKU:~/distributed-consensus-engine$ curl http://localhost:5002/pbft_test  
{"message":"TEST_PBFT","signature":"2dae36d5f48f8e951684e7867d462afdb51c4467a16c6bae12"}  
• (venv) (base) vamsh@DESKTOP-K8IJVKU:~/distributed-consensus-engine$ curl http://localhost:5003/pbft_test  
{"message":"TEST_PBFT","signature":"c51ee5b17f23c960c0ca64347fe2543dff1eb62d04e1fe56d0"}  
• (venv) (base) vamsh@DESKTOP-K8IJVKU:~/distributed-consensus-engine$ curl http://localhost:5004/pbft_test  
{"message":"TEST_PBFT","signature":"752ff63c984e8aef817ba1dcabb432897a711e5b6517df01df"}  
• (venv) (base) vamsh@DESKTOP-K8IJVKU:~/distributed-consensus-engine$ curl http://localhost:5005/pbft_test  
{"message":"TEST_PBFT","signature":"736652f814d73cb98f07ea2414fe3e424f6398257652bd0e42"}  
○ (venv) (base) vamsh@DESKTOP-K8IJVKU:~/distributed-consensus-engine$
```

Client

|

Primary (Leader)

|

PRE-PREPARE

|

Replicas

|

PREPARE

|

Replicas

|

COMMIT

|

Consensus

|

Ledger Commit

```
✓ Image distributed-consensus-engine-node1 Built
○ (venv) (base) vamsh@DESKTOP-K8IJVKU:~/distributed-consensus-engine$ sudo docker compose up
[+] up 6/6
✓ Network distributed-consensus-engine_default Created
✓ Container node4 Created
✓ Container node3 Created
✓ Container node1 Created
✓ Container node5 Created
✓ Container node2 Created
Attaching to node1, node2, node3, node4, node5
[]
```

```
● (venv) (base) vamsh@DESKTOP-K8IJVKU:~/distributed-consensus-engine$ curl http://localhost:5005/leader
{"leader_id":5}
● (venv) (base) vamsh@DESKTOP-K8IJVKU:~/distributed-consensus-engine$ curl -X POST http://localhost:5005/pbft_transaction \
-H "Content-Type: application/json" \
-d '{"txn":"PBFT_TEST_1"}'
{"mode":"PBFT","status":"committed"}
● (venv) (base) vamsh@DESKTOP-K8IJVKU:~/distributed-consensus-engine$ curl http://localhost:5005/ledger
{"leader_id":5,"ledger":[{"txn":"PBFT_TEST_1"}],"node_id":5}
○ (venv) (base) vamsh@DESKTOP-K8IJVKU:~/distributed-consensus-engine$ []
```



```
(venv) (base) vamsh@DESKTOP-K8IJVKU:~/distributed-consensus-engine$ curl http://localhost:5099/status
{"hostname":"5d7ba595301d","node_id":99,"role":"BYZANTINE"}
(venv) (base) vamsh@DESKTOP-K8IJVKU:~/distributed-consensus-engine$ curl -X POST http://localhost:5099/malicious_prepare \
-H "Content-Type: application/json" \
-d '{"target":"node1"}'
{"fake_txn":"TXN-100","status":"malicious"}
(venv) (base) vamsh@DESKTOP-K8IJVKU:~/distributed-consensus-engine$ curl -X POST http://localhost:5099/malicious_prepare \
-H "Content-Type: application/json" \
-d '{"target":"node2"}'
{"fake_txn":"TXN-999","status":"malicious"}
```

```
(venv) (base) vamsh@DESKTOP-K8IJVKU:~/distributed-consensus-engine$ sudo docker ps
CONTAINER ID   IMAGE                                COMMAND                  CREATED        STATUS        PORTS                               N
3c5228b37ee4   distributed-consensus-engine-node4   "python src/node.py"    About a minute Up About a minute 0.0.0.0:5004->5000/tcp, [::]:5004->5000/tcp n
ode4
6d7c46ddfd7c   distributed-consensus-engine-node3   "python src/node.py"    About a minute Up About a minute 0.0.0.0:5003->5000/tcp, [::]:5003->5000/tcp n
ode3
21f748895dbd   distributed-consensus-engine-node2   "python src/node.py"    About a minute Up About a minute 0.0.0.0:5002->5000/tcp, [::]:5002->5000/tcp n
ode2
b343b8113b90   distributed-consensus-engine-node1   "python src/node.py"    About a minute Up About a minute 0.0.0.0:5001->5000/tcp, [::]:5001->5000/tcp n
ode1
840f110755dd   distributed-consensus-engine-node5   "python src/node.py"    About a minute Up About a minute 0.0.0.0:5005->5000/tcp, [::]:5005->5000/tcp n
ode5
5d7ba595301d   distributed-consensus-engine-adversary "python src/adversar..." About a minute Up About a minute 0.0.0.0:5099->5000/tcp, [::]:5099->5000/tcp a
dversary
(venv) (base) vamsh@DESKTOP-K8IJVKU:~/distributed-consensus-engine$ curl http://localhost:5099/status
{"hostname":"5d7ba595301d","node_id":99,"role":"BYZANTINE"}
```

chaos_test.sh

This below figures shows the successful completion of the chaos testing process. The distributed cluster remained operational after multiple failures, recoveries, and network partitions.

Observation: The testing workflow completed without causing total system failure.

Outcome: The implementation demonstrated resilience, recovery capability, and operational continuity under realistic distributed-system fault scenarios.

Start cluster



Send transactions



Kill leader



[Verify re-election](#)



Restart leader



Partition network



Recover network



Continue transactions

```
(venv) (base) vamsih@DESKTOP-K81JVKU:~/distributed-consensus-engine$ sudo bash tests/chaos_test.sh
PBFY / PAXOS CHAOS TEST STARTED
=====

Current Containers
CONTAINER ID   IMAGE                                COMMAND                  CREATED        STATUS        PORTS                               NAMES
85802f040faf   distributed-consensus-engine-adversary  "python src/adversar..." 8 minutes ago  Up 8 minutes  0.0.0.0:5099->5000/tcp, [::]:5099->5000/tcp  adversary
db0b0274ce1b   distributed-consensus-engine-node3      "python src/node.py"      8 minutes ago  Up 8 minutes  0.0.0.0:5003->5000/tcp, [::]:5003->5000/tcp  node3
3928d2f8da2f   distributed-consensus-engine-node5      "python src/node.py"      8 minutes ago  Up 8 minutes  0.0.0.0:5005->5000/tcp, [::]:5005->5000/tcp  node5
6df1a593a62d   distributed-consensus-engine-node4      "python src/node.py"      8 minutes ago  Up 8 minutes  0.0.0.0:5004->5000/tcp, [::]:5004->5000/tcp  node4
6d427d3640dd   distributed-consensus-engine-node1      "python src/node.py"      8 minutes ago  Up 8 minutes  0.0.0.0:5001->5000/tcp, [::]:5001->5000/tcp  node1
fafd2f6a41bb   distributed-consensus-engine-node2      "python src/node.py"      8 minutes ago  Up 8 minutes  0.0.0.0:5002->5000/tcp, [::]:5002->5000/tcp  node2

Waiting for cluster stabilization...

Checking Leader
{"leader_id":5}

=====
TEST 1 : NORMAL TRANSACTIONS
=====
{"leader":5,"status":"committed"}
{"leader":5,"status":"committed"}

Leader Ledger
{"leader_id":5,"ledger":[{"txn":"NORMAL_TXN_1"}, {"txn":"NORMAL_TXN_2"}], "node_id":5}

=====
TEST 2 : LEADER CRASH
=====
[]
```

```
=====
TEST 2 : LEADER CRASH
=====
node5

Leader stopped

New Leader
{"leader_id":4}

=====
TEST 3 : TRANSACTION AFTER FAILURE
=====
{"status":"failed"}
{"leader_id":4,"ledger":[], "node_id":4}

=====
TEST 4 : LEADER RECOVERY
=====
node5

Leader Status
{"leader_id":5}

=====
TEST 5 : BYZANTINE NODE
=====
{"hostname":"85802f040faf", "node_id":99, "role":"BYZANTINE"}
```

```

=====
TEST 5 : BYZANTINE NODE
=====
{"hostname":"85802f040faf","node_id":99,"role":"BYZANTINE"}

{"fake_txn":"TXN-100","status":"malicious"}

{"fake_txn":"TXN-999","status":"malicious"}

=====
TEST 6 : NETWORK PARTITION
=====

Node3 isolated

Remaining Cluster
{"leader_id":5}

=====
TEST 7 : RECOVER PARTITION
=====

Node3 rejoined cluster
{"hostname":"db0b0274ce1b","leader_id":5,"ledger_size":0,"node_id":3,"status":"alive"}

```

```

=====
TEST 8 : FINAL TRANSACTION
=====
{"status":"failed"}
{"leader_id":5,"ledger":[],"node_id":5}

=====
CHAOS TEST COMPLETED
=====
○ (venv) (base) vamsh@DESKTOP-K8IJVKU:~/distributed-consensus-engine$ █

```